

QUESTION 5(a)

Question:

To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

```
GNU nano 8.7.1 SKILL5a.c *
#include <stdio.h>
#include <string.h>

#define MAX 200

int main()
{
    char input[MAX];
    char result[MAX];
    int i = 0, j = 0;

    printf("==== Single Quote Parser ==== \n");
    printf("Enter text inside single quotes: ");

    fgets(input, MAX, stdin);
    input[strcspn(input, "\n")] = '\0';

    /* Check for single quotes */
    if (input[0] != '\'' ||
        input[strlen(input) - 1] != '\')
    {
        printf("Error: Input must be enclosed in single quotes.\n");
        return 1;
    }

    /* Remove the single quotes */
    i = 1;

    while (input[i] != '\0' &&
           input[i] != '\') &&
           j < MAX - 1)
    {
        result[j++] = input[i++];
    }

    result[j] = '\0';
}
```

```

    /* Validate closing quote */
    if (input[i] != '\')
    {
        printf("Error: Missing closing single quote.\n");
        return 1;
    }

    printf("\nOriginal Input : %s\n", input);
    printf("Parsed Content : %s\n", result);

    printf("\nSingle quotes preserve the literal content.\n");
    printf("Variable expansion is ignored inside single quotes.\n");

    return 0;
}

```

Output

```

sri_dhruthi@Dhruthi:~$ nano skill5a.c
sri_dhruthi@Dhruthi:~$ gcc skill5a.c -o skill5a
sri_dhruthi@Dhruthi:~$ ./skill5a
===== Single Quote Parser =====
Enter text inside single quotes: 'Hello World'

Original Input : 'Hello World'
Parsed Content : Hello World

Single quotes preserve the literal content.
Variable expansion is ignored inside single quotes.
sri_dhruthi@Dhruthi:~$ ./skill5a
===== Single Quote Parser =====
Enter text inside single quotes: 'Hello $USER'

Original Input : 'Hello $USER'
Parsed Content : Hello $USER

Single quotes preserve the literal content.
Variable expansion is ignored inside single quotes.
sri_dhruthi@Dhruthi:~$

```

QUESTION 5(b)

Question:

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 200

int main()
{
    char input[MAX];
    char result[MAX];
    char variable[MAX];

    int i = 0;
    int j = 0;

    printf("==== Double Quote Parser ====\\n");
    printf("Enter text inside double quotes: ");

    fgets(input, MAX, stdin);
    input[strcspn(input, "\\n")] = '\\0';

    /* Check opening and closing double quotes */
    if (input[0] != '"' ||
        input[strlen(input) - 1] != '"')
    {
        printf("Error: Input must be enclosed in double quotes.\\n");
        return 1;
    }

    i = 1;

    while (input[i] != '\\0' &&
           input[i] != '"' &&
           j < MAX - 1)
    {

```

```
/* Variable expansion */
if (input[i] == '$')
{
    int k = 0;
    i++;

    while ((input[i] >= 'A' && input[i] <= 'Z') ||
           (input[i] >= 'a' && input[i] <= 'z') ||
           (input[i] >= '0' && input[i] <= '9') ||
           input[i] == '_')
    {
        if (k < MAX - 1)
            variable[k++] = input[i];

        i++;
    }

    variable[k] = '\0';

    char *value = getenv(variable);

    if (value != NULL)
    {
        for (int x = 0;
             value[x] != '\0' && j < MAX - 1;
             x++)
        {
            result[j++] = value[x];
        }
    }
    else
    {
        result[j++] = input[i++];
    }
}
```

```

    result[j] = '\0';

    /* Validate closing quote */
    if (input[i] != '"')
    {
        printf("Error: Missing closing double quote.\n");
        return 1;
    }

    printf("\nOriginal Input : %s\n", input);
    printf("Parsed Content : %s\n", result);

    printf("\nDouble quotes preserve spaces.\n");
    printf("Variable expansion is allowed inside double quotes.\n");

    return 0;
}

```

Output

```

sri_dhruthi@Dhruthi:~$ nano skill5b.c
sri_dhruthi@Dhruthi:~$ gcc skill5b.c -o skill5b
sri_dhruthi@Dhruthi:~$ ./skill5b
===== Double Quote Parser =====
Enter text inside double quotes: "Hello World Welcome"

Original Input : "Hello World Welcome"
Parsed Content : Hello World Welcome

Double quotes preserve spaces.
Variable expansion is allowed inside double quotes.
sri_dhruthi@Dhruthi:~$ ./skill5b
===== Double Quote Parser =====
Enter text inside double quotes: "Hello $USER"

Original Input : "Hello $USER"
Parsed Content : Hello sri_dhruthi

Double quotes preserve spaces.
Variable expansion is allowed inside double quotes.
sri_dhruthi@Dhruthi:~$ |

```